# MapReduce Workflow – Detailed 13 Mark Answer

**Introduction**
MapReduce is a programming model introduced by Hadoop for processing large volumes of data in a distributed environment. It divides a task into smaller sub-tasks and processes them in parallel across multiple nodes in a cluster. The workflow of MapReduce explains how data moves from input files stored in HDFS to the final output generated after reduction.

## 1. Input Data Storage in HDFS

Data is stored in Hadoop Distributed File System (HDFS). Large files are split into blocks and distributed across nodes. Replication provides fault tolerance and reliability. This phase plays a critical role in achieving scalability, parallelism, and fault tolerance in Hadoop MapReduce processing.

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## 2. Input Splitting

InputFormat divides files into logical InputSplits. Each split is assigned to a Mapper. RecordReader converts raw data into key-value pairs. This phase plays a critical role in achieving scalability, parallelism, and fault tolerance in Hadoop MapReduce processing.

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## 3. Mapper Phase

The Mapper processes input records and produces intermediate key-value pairs. It performs filtering, transformation, and extraction of useful information. This phase plays a critical role in achieving scalability, parallelism, and fault tolerance in Hadoop MapReduce processing.

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## 4. Combiner Phase

The Combiner acts as a mini-reducer and performs local aggregation before data is transferred across the network. This phase plays a critical role in achieving scalability, parallelism, and fault tolerance in Hadoop MapReduce processing.

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## 5. Partitioning

Partitioner determines which reducer receives a specific key. Same keys always go to the same reducer. This phase plays a critical role in achieving scalability, parallelism, and fault tolerance in Hadoop MapReduce processing.

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## 6. Shuffle Phase

Shuffle transfers intermediate data from mappers to reducers and groups identical keys together. This phase plays a critical role in achieving scalability, parallelism, and fault tolerance in Hadoop MapReduce processing.

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## 7. Sorting Phase

Keys are automatically sorted before reaching reducers, ensuring organized processing. This phase plays a critical role in achieving scalability, parallelism, and fault tolerance in Hadoop MapReduce processing.

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## 8. Reducer Phase

Reducers aggregate grouped values and generate the final result. This phase plays a critical role in achieving scalability, parallelism, and fault tolerance in Hadoop MapReduce processing.

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## 9. Output Phase

The final output is written back to HDFS using OutputFormat and RecordWriter. This phase plays a critical role in achieving scalability, parallelism, and fault tolerance in Hadoop MapReduce processing.

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# Complete MapReduce Workflow Example

Input File:
Cat Dog Cat
Dog Lion

Mapper Output:
<Cat,1> <Dog,1> <Cat,1> <Dog,1> <Lion,1>

Shuffle and Sort Output:
<Cat,[1,1]> <Dog,[1,1]> <Lion,[1]>

Reducer Output:
Cat 2
Dog 2
Lion 1

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# Advantages of MapReduce Workflow

1. Parallel processing.
2. High scalability.
3. Fault tolerance through HDFS replication.
4. Efficient processing of huge datasets.
5. Distributed computing support.
6. Cost-effective architecture.
7. Automatic load balancing.
8. Reliable execution in clusters.


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# Conclusion

MapReduce workflow is the foundation of Hadoop data processing. By dividing computation into mapping and reducing phases and using shuffle and sort mechanisms, Hadoop can efficiently process massive datasets across distributed systems. The workflow ensures scalability, fault tolerance, reliability, and high performance, making it one of the most important concepts in Big Data technologies.

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